

GraphMARL: Graph Neural Multi-Agent Reinforcement Learning for Fully Decentralized Peer-to-Peer Hospital Edge Orchestration

1. Problem Statement & Baseline

While recent literature, including the baseline paper ("Task prioritization and distributed deep reinforcement learning for healthcare management in Cloud-Edge environments"), introduces effective methods for prioritizing healthcare tasks using a Health Severity Index (HSI) and Distributed Deep Reinforcement Learning (DRL), it suffers from several critical limitations:

- **Centralized Orchestration:** It relies on a logically centralized orchestrator or global state knowledge for task offloading.
- **Vertical Offloading Only:** It focuses on Device-to-Edge and Edge-to-Cloud offloading, missing the massive potential of horizontal Edge-to-Edge (Peer-to-Peer) collaboration across hospital departments.
- **Lack of Graph Intelligence:** It ignores the inherent graph-like topology of interconnected hospital departments.
- **Simulation-Only Validation:** It lacks validation on real-world hardware under realistic communication failures.
- **Single Point of Failure:** Centralized scheduling makes the system vulnerable to node crashes and network partitions.

2. Proposed Solution: How We Will Solve These Problems

Our proposed framework, **GraphMARL**, completely eliminates the central scheduler. Every hospital department (e.g., ICU, Radiology, ER) acts as an autonomous, intelligent edge agent capable of Peer-to-Peer collaboration. We solve the gaps in the base paper through the following methodologies:

- **Fully Decentralized Peer-to-Peer Orchestration:** Nodes make decisions independently. If a node fails, the rest of the hospital continues to function, routing tasks horizontally to other departments.
- **Dynamic Hospital Graph Representation:** The hospital is modeled as a dynamic graph $G = (V, E)$, where vertices are departments and edges are communication links. Node and edge features update continuously.
- **Hybrid State Fusion (SCNN + GNN):** We utilize a Stacked CNN (SCNN) to compress the local department's high-dimensional state (compute, energy) and fuse it with a Graph Neural Network (GNN) that aggregates compact neighbor embeddings.
- **Uncertainty-Aware Decentralized Decision Making:** We integrate an Uncertainty-Aware Multi-Agent Proximal Policy Optimization (MAPPO) algorithm. The agent calculates the variance of its Q-value predictions for neighboring nodes. If neighbor data is stale or network congestion is high, the agent penalizes offloading to that neighbor, preventing catastrophic delays for critical medical tasks.
- **Human-in-the-Loop (HITL) Override:** Unlike purely autonomous systems, clinicians can inject real-time priority overrides which dynamically propagate through the graph, forcing neighboring nodes to clear their queues for emergency tasks.
- **Real Hardware Validation:** We will deploy the system on a testbed of NVIDIA Jetson boards (Nano, Xavier, Orin) simulating real hospital departments, injecting realistic network delays, packet loss, and

node crashes.

3. Core Methodologies Introduced

A. Dynamic Hospital Graph Constructor (DHGC)

Maintains the evolving graph of departments and communication links using local observations. Features include CPU/GPU utilization, queue length, and network bandwidth.

B. Hybrid Perception Network (SCNN + NAGEN)

Combines two perception layers:

- 1. **SCNN**: Compresses high-dimensional local state parameters.
- 2. **Neighbor-Aware Graph Embedding Network (NAGEN)**: Utilizes GNNs (like GraphSAGE) to produce node embeddings from local neighborhoods. The fused embedding (`[Local SCNN | Neighbor GNN]`) captures both local capacity and surrounding hospital congestion without global knowledge.

C. Health Severity Priority Encoder (HSPE) & HITL Override

Extends the Health Severity Index (HSI). HSI is integrated directly into the MARL state space. Furthermore, the **HITL Override Protocol** allows clinicians to manually flag tasks as emergencies, instantly elevating the task's HSI and propagating "Emergency Override Embeddings" to neighbors.

D. Uncertainty-Aware GraphMARL Scheduler (UA-GMS)

A multi-agent policy where each node learns independently. To prevent risky offloading due to stale neighbor information, the Q-learning objective incorporates variance-based uncertainty estimation. High prediction variance mathematically penalizes the action. The Reward function is multi-objective: $R = \alpha U - \beta L - \gamma E - \delta D - \eta C + \lambda P - \mu \text{Var}(Q)$ Where U is utilization, L is latency, E is energy, D is deadline misses, C is communication cost, P is critical-task completion, and $\text{Var}(Q)$ is the prediction uncertainty penalty.

E. Adaptive Neighbor Negotiation Protocol (ANNP)

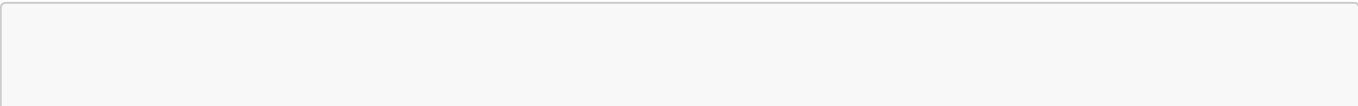
A lightweight protocol for exchanging compact graph embeddings and predicted finish times instead of full resource tables, drastically reducing communication overhead.

F. Dynamic Failure Recovery Module (DFRM)

Handles node failures (e.g., heartbeats drop), stale information, and network partitions. The graph dynamically updates, and nodes route around failures autonomously.

4. Workflows & Architecture Diagrams

System Architecture Workflow



```

graph TD
    A[Medical IoT Devices] -->|Raw Vitals & Images| B(Hybrid Transformer-BiLSTM Feature Extraction)
    B -->|Task Metadata| C[Health Severity Estimator]
    C -->|Task ID, HSI, Compute Needs| D{Task Characterization Layer}
    D -->|Incoming Task| E[Local Department Edge Node e.g., ICU, ER]

    subgraph Clinician Feedback
        Z[Doctor/System Admin] -->|HITL Override Alert| C
    end

    subgraph Decentralized Graph Intelligence
        E --> F[Neighbor Discovery via ANNP]
        F -->|Exchange Embeddings| G[Distributed Graph Construction]
        G -->|Local State| H1[SCNN Feature Extractor]
        G -->|Neighbor States| H2[GNN Encoder GAT/GraphSAGE]
        H1 --> I{State Fusion}
        H2 --> I
    end

    subgraph Multi-Agent Decision Engine
        I -->|Fused State| J[Uncertainty-Aware MARL MAPPO]
        J -->|Calculate Q-Variance| K{Local Offloading Decision Engine}
    end

    K -->|Low Uncertainty| L[Local Execution Queue]
    K -->|Offload Horizontal| M[Neighbor Edge Node]
    K -->|High Uncertainty| N[Retry / Escalate]

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    style J fill:#fff9c4,stroke:#fbc02d
    style H2 fill:#e8f5e9,stroke:#2e7d32
    style Z fill:#ffcdd2,stroke:#c62828

```

GraphMARL Decision Flow (Perception to Execution)

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flowchart LR
    subgraph Perception Phase
        A1[Local State Features] --> SCNN[Stacked CNN]
        A2[Neighbor Embeddings] --> GNN[Graph Neural Network]
        SCNN --> Fused[Fused State Representation]
        GNN --> Fused
    end

    subgraph Decision Phase
        Fused --> MARL[UA-MARL Policy]
        T[Task Priority / HSI / HITL] --> MARL
        MARL -->|Evaluate Q-Variance| D{Decision Action}
    end

    subgraph Execution Phase

```

```

D -->|Execute Local| E1[Local GPU/CPU]
D -->|Edge-to-Edge Offload Left| E2[Neighbor Dept A]
D -->|Edge-to-Edge Offload Right| E3[Neighbor Dept B]
D -->|Delay| E4[Wait Queue]
end

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style MARL fill:#fff9c4,stroke:#fbc02d
style SCNN fill:#e8f5e9,stroke:#2e7d32

```

5. Our 15 Contributions

1. **Fully Decentralized Architecture:** Propose the first fully decentralized peer-to-peer hospital edge orchestration framework without any central scheduler.
2. **Horizontal Edge-to-Edge Collaboration:** Shift the paradigm from vertical (Device-to-Cloud) offloading to lateral Peer-to-Peer department collaboration.
3. **Dynamic Graph Modeling:** Model the hospital infrastructure as a dynamic graph where departments are nodes and communication links are graph edges.
4. **Hybrid Perception Module (SCNN + GNN):** Introduce a dual-perception architecture combining SCNNs for high-dimensional local states and GNNs for neighborhood-aware resource representation.
5. **GraphMARL Framework:** Develop a GraphMARL scheduling framework by integrating Graph Neural Networks with Multi-Agent Reinforcement Learning.
6. **Uncertainty-Aware Decentralized Policy:** Implement an uncertainty estimation mechanism that calculates Q-value variance to prevent high-risk offloading in dynamic, partially observable network conditions.
7. **Human-in-the-Loop (HITL) Override:** Design a closed-loop feedback mechanism allowing clinicians to inject real-time priority overrides that instantly propagate through the graph.
8. **Lightweight Negotiation Protocol:** Design a lightweight neighbor-to-neighbor negotiation protocol (ANNP) using graph embeddings instead of complete resource exchange.
9. **HSI-Integrated Scheduling:** Develop deadline-aware and Health Severity Index (HSI)-based intelligent task scheduling for critical healthcare applications.
10. **Congestion-Aware Offloading:** Introduce adaptive congestion-aware task offloading using graph-based neighborhood intelligence.
11. **Dynamic Failure Recovery:** Develop dynamic failure recovery mechanisms for node failures, stale information, and network partitioning.
12. **Fairness-Aware Workloads:** Incorporate fairness-aware decentralized scheduling to balance workloads among hospital departments.
13. **Low Communication Overhead:** Minimize communication overhead through localized graph aggregation and neighbor-only information exchange.
14. **Real Hardware Validation:** Validate the proposed framework on a real Jetson-based hospital edge testbed in addition to extensive simulations.
15. **Comprehensive Superiority:** Demonstrate improved latency, deadline satisfaction, resource utilization, scalability, robustness, and fault tolerance over existing healthcare edge scheduling approaches.